Zhalok Rahman

Software Engineer 2 at Datatree Technology
rahmanzhalok@gmail.com
phone +8801716922067
LinkedIn: https://www.linkedin.com/in/zhalok-rahman/
Github: https://github.com/zhalok
Video Introduction: https://www.loom.com/share/c5ab46a4737244edafd9f052555767a1

Skills

Golang, Python (FastAPI, Asyncio, Pytorch, Pandas), Node JS (Nest JS, Express JS, Koa), JavaScript/TypeScript (React JS, Next JS, PostgreSQL, MongoDB, Redis (Cache, Pub/Sub, Streams), RabbitMQ, WebSocket, SSE, Docker, AWS (ECS, CloudWatch, SQS, Kinesis, Lambda, OpenSearch, DynamoDB), Grafana, Prometheus, CI/CD (Github Actions, CodePipeline

Experience

Kinetik Healthcare *Software Engineer 2*
2025-04-01 – Present

- Built voice AI agent for customer support
- Integrated voice AI agent with existing callcenter for handling fallback
- Built metrics consuming pipeline for monitoring AI agent
- Containerized and deployed AI services to ECS
- Managed automated pipeline for deploying AI agent
- Built custom VoIP tool for testing voice AI agent
- Integrated AI agent with core system data source
- Built dashboards for monitoring AI agent performance
- Set alerting for detecting failures and downtime in AI agent
- Integrated langfuse for tracing AI agent activity across multiple services

PropertyFinder *Software Engineer*
2024-10-01 – 2026-03-31

- Worked as on-call engineer to provide technical support to the customer raised issues
- Investigated and fixed root cause of critical production incidents
- Mitigated production incidents with scripting
- Conducted load testing to ensure system stability before new release
- Monitored metrics, logs and traces for detecting failures
- Modernized legacy B2B service with golang and AWS services

Learnyx AI *Fullstack Engineer*
2023-10-01 – 2025-02-01

- Built AI

Innovext IT *Fullstack Developer*
2022-12-01 – 2023-10-01

- Built REST apis with Nest JS, Express JS and FastAPI (for deploying fine-tuned model)
- Implemented payment and monetary transaction features by integrating third party payment gateways (stripe and paypal)
- Built client side application with Next JS, used Redux for efficient data fetching and dataflow
- Built an internal monetization system with credits, ensured smooth credit transfer with mongodb sessions
- Built server with FastAPI for inferencing fine-tuned object detection model in onnx format
- Created scripts for data preprocessing and training models

Projects

Remote Browser Control
Remote-controlled Playwright browser session streamed to a React client over WebSocket.

- Built FastAPI app for receiving commands for playwright to apply on browser session via websocket and emit current snapshot of the browser
- Created a front end app with react to render the remote browser session and also send commands via websocket
- Demo: https://bit.ly/4wDD7om
- Github: https://github.com/zhalok/remote-browser-control

Tech: Playwright, Python, FastAPI, WebSocket

AI Product Search

Semantic product search over CSV catalogs using SBERT embeddings, Qdrant and an async RabbitMQ indexing pipeline. https://github.com/zhalok/ai-product-search

- Built async vector indexing pipeline from csv file with RabbitMQ and Qdrant
- Built inferencing and searching pipeline with FastAPI
- Used finetuned self hosted SBERT model for generating embedding
- Github:

Tech: FastAPI, VectorDB (Qdrant), RabbitMQ, SBERT

PostgreSPECTOR

Observability toolkit for PostgreSQL that exposes execution stats and container metrics to Prometheus and Grafana, with logs shipped via Loki.

- Demo: https://youtu.be/w2OCn0X3bRw
- Github: https://bit.ly/4y8AFXX

Tech: Golang, Prometheus, Grafana, PostgreSQL, Loki, Promtail, Docker

Publications

Optical Mark Recognition with Object Detection and Clustering

2024 6th International Conference on Electrical Engineering and Information and Communication Technology (ICEE-ICT) *May 2024*

`https://doi.org/10.1109/ICEEICT62016.2024.10534561`

- Written **Python** scripts for generating synthetic training data
- Built training scripts in **Python** with ultralytics for training lightweight YOLO models
- Implemented clustering based unsupervised algorithm for column detection and serialization of answers
- Exported finetuned model in Onnx format for inferencing via API
- demo: https://www.youtube.com/watch?v=BiEybE5qQqw

Detecting Communal Violence in Texts

Association for Computational Linguistics *May 2024*

`https://aclanthology.org/2023.banglalp-1.25/`

- Curated and annotated a large-scale dataset of violence-related texts aggregated from multiple heterogeneous sources
- Benchmarked multiple pretrained BERT variants in **Python** to identify the optimal backbone for communal violence detection
- Applied domain-adaptive pretraining and fine-tuning in **Python** to boost classification accuracy on low-resource Bangla text

Certifications

AWS Certified Cloud Practitioner

AWS

https://www.credly.com/badges/201260d3-12e0-4d22-a49f-4e64d68b348e

Deep dive on AWS services

Optimizing MongoDB Performance with Tuning Tools

MongoDB

https://www.credly.com/badges/e861b7b4-028a-47d4-aed6-7e586d4d95da

Learned about diagnosing and tuning MongoDB performance using database tooling

Supervised Machine Learning: Regression and Classification

Coursera

https://coursera.org/share/0ef4defbf44ee7d868c52c7e1a3bb94f

Supervised learning from scratch, learning about mathematical algorithms working as backbone of machine learning models

Education

Shahjalal University of Science and Technology

Bachelors of Science in Computer Science

2019-01-01 – 2024-04-01